

University Interscholastic League
2022 – 2023 Junior High Number Sense Test A

Contestant's Number _____

Final		
2 nd		
1 st		
	Score	Initials

**Read Directions Carefully
Before Beginning Test**

**Do Not Unfold This Sheet
Until Told to Begin**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.

Stop – Wait for Signal!

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| <p>(1) $23 + 51 =$ _____</p> <p>(2) $18 \times 5 =$ _____</p> <p>(3) $79 - 26 =$ _____</p> <p>(4) $1203 \div 3 =$ _____</p> <p>(5) $\frac{3}{8} + \frac{1}{4} =$ _____</p> <p>(6) $24 \div 0.5 =$ _____</p> <p>(7) $40 \times 0.25 =$ _____</p> <p>(8) $24^2 =$ _____</p> <p>(9) $11 \times 5 \times 2 =$ _____</p> <p>*(10) $21 + 2022 + 23 + 2023 =$ _____</p> <p>(11) $1.75 \times 8 =$ _____</p> <p>(12) The mode of 2, 6, 8, 2, 6 and 2 is _____</p> <p>(13) Which is larger: $\frac{6}{11}$ or 0.55? _____</p> <p>(14) $15 \times 35 =$ _____</p> <p>(15) $38 \times 78 =$ _____</p> <p>(16) $97 - 12 - 18 =$ _____</p> <p>(17) $43^2 =$ _____</p> <p>(18) $9 + 14 + 19 + 24 + 29 =$ _____</p> <p>(19) MCMLI = _____ (Arabic Numeral)</p> | <p>*(20) $111.111\% \times 899 =$ _____</p> <p>(21) $0.75 \times 16 - 0.5 \times 16 =$ _____</p> <p>(22) $111 \times 135 =$ _____</p> <p>(23) If $f(x) = 10 - 3x^2$, then $f(-2) =$ _____</p> <p>(24) $8\frac{1}{3} \times 8\frac{2}{3} =$ _____ (mixed number)</p> <p>(25) $7\frac{3}{4} - 4\frac{5}{6} =$ _____</p> <p>(26) $29^2 + 29 =$ _____</p> <p>(27) The cube root of -125 is _____</p> <p>(28) If n is to 20 as 12 is to 4, then $n =$ _____</p> <p>(29) If $12 - 4x$ is 4 then $x =$ _____</p> <p>*(30) $2023 + 2022 \times 4 =$ _____</p> <p>(31) $93 \times 92 =$ _____</p> <p>(32) The product of the two largest prime numbers less than 20 is _____</p> <p>(33) If $0.025 + 0.075 = n$, the $n^{-1} =$ _____</p> <p>(34) 18 is divisible by how many whole numbers? _____</p> <p>(35) 1 square mile = _____ acres</p> <p>(36) 4 percent = _____ (common fraction)</p> <p>(37) The total cost of item, that costs \$10, with a sales tax of $8\frac{1}{2}\%$ is \$ _____</p> |
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- (38) $231 \times 101 =$ _____
- (39) The area of an equilateral triangle with an altitude of 6-cm is $a\sqrt{3}$ and $a =$ _____ cm^2
- *(40) $125\sqrt{14400} =$ _____
- (41) 63 (base 8) = _____ (base 2)
- (42) $5^3 + 3^3 =$ _____
- (43) The GCF of 6, 12 and 18 is _____
- (44) What is the area of a trapezoid with base 17, 16 and height 20? _____
- (45) 23 (base 5) $\times 2$ (base 5) = _____ (base 5)
- (46) If $24 \div 3 = n \div 2$, then $n =$ _____
- (47) What is the length of an edge of a cube with surface area 864? _____
- (48) $1.5 \text{ m}^2 =$ _____ cm^2
- (49) If $24 - 3x < 18$, then $2x >$ _____
- *(50) $55^3 =$ _____
- (51) In the sequence: 1, 4, a , 16, 25, b , 49, ...
 $2a - b =$ _____
- (52) What is the 6th term in the sequence 2, 4, 6, ...? _____
- (53) What is the sum of the interior angles for a convex polygon with 5 sides? _____ $^\circ$
- (54) What is the product of the LCM and GCD of 18 and 20? _____
- (55) What is the length of the side of a rectangle with perimeter 44 and width 8? _____
- (56) What is the diameter of a circle with an area of 576π ? _____
- (57) If set $A = \{1, 2, 3, \dots, 19\}$ and set $B = \{3, 6, 9, \dots, 21\}$, then the number of elements in $A \cap B$ is _____
- (58) $(13^2 + 7 \times 6) \div 5$ has a remainder of _____
- (59) The number of faces for a cube is _____
- *(60) $749^2 \times 101 \div 251 =$ _____
- (61) $(2^{-1} + 2^{-2})^{-1} =$ _____
- (62) 440 yards = _____ mile
- (63) 60 feet/sec = _____ yards/min
- (64) 18% of 18 = _____
- (65) $5^7 \div 7$ has a remainder of _____
- (66) If the volume of a right circular cylinder, with radius 4 and length 12 is $k\pi$, what is k ? _____
- (67) The sum of the positive divisors of 12 is _____
- (68) $6! =$ _____
- (69) $6\frac{3}{4} \div \frac{1}{8} =$ _____
- *(70) $350\pi^3 =$ _____
- (71) $21^2 + 3^2 =$ _____
- (72) $62\frac{1}{2}\% =$ _____ (common fraction)
- (73) $0.4666\dots =$ _____ (common fraction)
- (74) The volume of a right circular cone with height 9 and radius 4 is $k\pi$, and $k =$ _____
- (75) The sum of the 4th and 5th triangular numbers is _____
- (76) $18^2 \times 25 =$ _____
- (77) If the shortest distance between the points (0, 3) and (x , 0) is 5, what is x ? _____
- (78) $\frac{5!}{3!} - 15 =$ _____
- (79) $\left(18\frac{1}{2}\right)^2 - \left(6\frac{1}{2}\right)^2 =$ _____
- *(80) 25 miles = _____ yards

2022 – 2023 University Interscholastic League Junior High Number Sense Test A – Key

(1) 74	*(20) 949 – 1048	(38) 23331	(59) 6
(2) 90	(21) 4	(39) 12	*(60) 214455 – 237028
(3) 53	(22) 14985	*(40) 14250 – 15750	(61) $1\frac{1}{3}; \frac{4}{3}$
(4) 401	(23) -2	(41) 110011	(62) $\frac{1}{4}; .25$
(5) $\frac{5}{8}; .625$	(24) $72\frac{2}{9}$	(42) 152	(63) 1200
(6) 48	(25) $2\frac{11}{12}; \frac{35}{12}$	(43) 6	(64) $3.24; 3\frac{6}{25}; \frac{81}{25}$
(7) 10	(26) 870	(44) 330	(65) 5
(8) 576	(27) -5	(45) 101	(66) 192
(9) 110	(28) 60	(46) 16	(67) 28
*(10) 3885 – 4293	(29) 2	(47) 12	(68) 720
(11) 14	*(30) 9606 – 10616	(48) 15000	(69) 54
(12) 2	(31) 8556	(49) 4	*(70) 10310 – 11394
(13) $.55; \frac{11}{20}$	(32) 323	*(50) 158057 – 174693	(71) 450
(14) 525	(33) 10	(51) -18	(72) $\frac{5}{8}$
(15) 2964	(34) 6	(52) 12	(73) $\frac{7}{15}$
(16) 67	(35) 640	(53) 540	(74) 48
(17) 1849	(36) $\frac{1}{25}$	(54) 360	(75) 25
(18) 95	(37) 10.85	(55) 14	(76) 8100
(19) 1951		(56) 48	(77) 4
		(57) 6	(78) 5
		(58) 1	(79) 300
			*(80) 41800 – 46200

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.